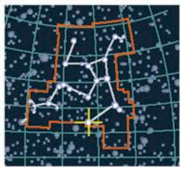
ECLIPSING BINARY

Alpha Herculis



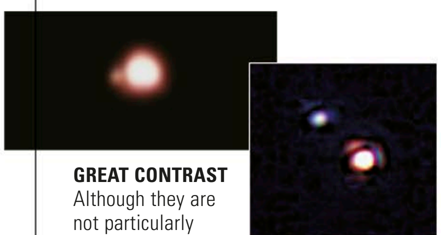
HERCULES

DISTANCE FROM SUN 382 light-years

MAGNITUDE 3

SPECTRAL TYPE M5

Alpha (α) Herculis, or Ras Algethi, is a cool red supergiant star that varies in brightness by almost one magnitude over a period of about 128 days. It is a complex star system with a much smaller companion that is itself a binary, consisting of a giant and a Sun-like star. There is a strong wind of stellar material blowing from the star, which reaches and engulfs its companions. Alpha Herculis is wider than the orbit of Mars. The outer atmosphere of the supergiant is slowly being removed, and the star will eventually become a white dwarf.



GREAT CONTRAST
Although they are not particularly bright, the great contrast in size and color of the stars that make up Alpha Herculis allow them to be separated easily through a telescope.

PULSATING VARIABLE

Mira

CETUS

DISTANCE FROM SUN 418 light-years

MAGNITUDE 3

SPECTRAL TYPE M7

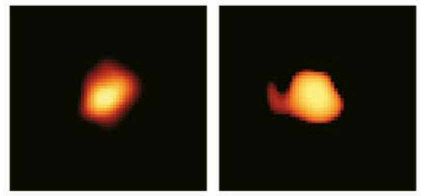
Omicron (ο) Ceti, better known as Mira, is among the best known of all variable stars. At its brightest, it reaches 2nd magnitude, and at its faintest, it drops to 10th—far too faint for the naked eye. It undergoes this variation with a period of 330 days. Therefore, an observer can find Mira when it is at its brightest and over a period of time watch it completely disappear. Although Mira is one of the coolest stars visible in the sky, with a temperature of just 3,600°F (2,000°C), it is at least 15,000 times more luminous than the Sun. Internal changes in the star have left it so distended that the Hubble Space Telescope has revealed that it is not perfectly spherical (right). The variation in Mira's magnitude is caused by pulsations that cause temperature changes and therefore changes in the star's luminosity. Furthermore, Mira is shedding material from its outer layers in the form of a stellar wind. In the future, Mira will lose its outer structure and be left as a small white dwarf. In this way, Mira represents the future of the Sun.

EXPLORING SPACE

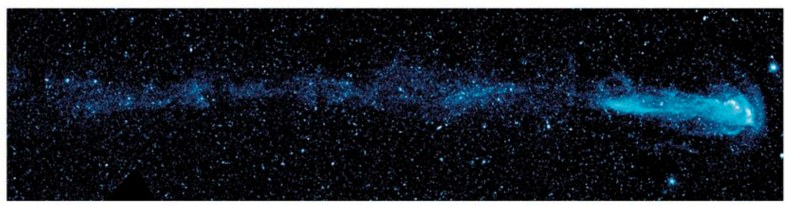
WONDERFUL MIRA

When German astronomer David Fabricius discovered Mira in 1596, it was the first long-period variable star to be recognized. In 1642, Johannes Hevelius named the star Mira, meaning "wonderful." It has become the most famous long-period pulsating variable in the sky and one of the most popular stars for amateur astronomers. The American Association of Variable Star Observers has received more than 50,000 observations of Mira by over 1,600 observers.

THE ORIGINAL MIRA
Easily recognized in the night sky, Mira lends its name to a type of long-period variable, of which thousands are known.



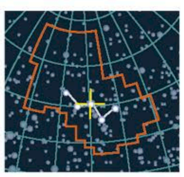
DISTORTED SHAPE
Enhancement of Hubble's Mira images reveals the star's asymmetrical atmosphere in visible (left) and ultraviolet light (right).



GLOWING TAIL
Mira is shedding gas as it moves through space, producing a tail 13 light-years long that shows up at ultraviolet wavelengths, as seen in this image from NASA's Galaxy Evolution Explorer.

IRREGULAR VARIABLE

Gamma Cassiopeiae



CASSIOPEIA

DISTANCE FROM SUN 613 light-years

MAGNITUDE 2.4

SPECTRAL TYPE B0

A hot blue star with a surface temperature of 45,000°F (25,000°C), Gamma (γ) Cassiopeiae is about 70,000 times more luminous than the Sun. It is a variable star with unpredictable changes in magnitude. Astronomers have observed it as bright as 1st magnitude and as faint as 3rd magnitude. It may have been fainter in ancient times, which might explain its lack of a common name. Gamma Cassiopeiae is a Be star (see panel, right), rotating at more than 625,000 mph (1 million kph) at its equator and shedding material from its surface. The ejected material forms a surrounding disk, and it is the disk that makes varying and unpredictable emissions. Gamma Cassiopeiae may also be transferring material to an undiscovered dense companion star.

NAMELESS STAR
Pictured here with the red-colored emission nebula IC 63, Gamma Cassiopeiae is among the most prominent stars in the sky that carries no common name.

EXPLORING SPACE

THE FIRST "BE" STAR

When, in 1866, Father Angelo Secchi, director of the Vatican Observatory and scientific advisor to Pope Pius IX, studied the spectrum of Gamma Cassiopeiae, he discovered that the star emitted light at particular wavelengths associated with hydrogen emission (see p.35). He is therefore credited with the discovery of the first Be star—a star of spectral type B but with "e" for emission. Be stars are characterized by their high rotation speeds, high surface temperatures, and strong stellar winds focused into equatorial disks.

CENTRAL STAR
Gamma Cassiopeiae, the brightest star in this image, is the central star in the distinctive "W" of Cassiopeia (see p.357).